

VTHACKS 14 · SEPTEMBER 2026 · IMPIRICUS TRACK

Aperta

Judge briefing. Everything we might be asked, and what we say.

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Live: **aperta.health** · also aniketh.net/aperta

Code: github.com/ankthba/VTHacks14

Educational demo. Not medical advice. No real patient data was used.

Read the first two pages if you read nothing else.

If you only read one page

The answer to "what is it?"

A clinician pastes the note they already wrote. Aperta turns it into what the patient still understands at home: where on their body, one plain sentence, each medicine and how to take it, what to do next, and how to actually do it. In their language, read aloud, one idea per screen, printed on one sheet. Then the clinician turns the screen around.

The three sentences that do the work

1. **The moment.** The most valuable screen in medicine is the one a doctor turns toward a patient, and nobody owns it. A rep is never in that room.
2. **The problem.** Patients forget most of what they are told in a visit. The fix everyone agrees on is teach-back, and a fifteen-minute slot does not allow it, so the explanation has to become an artifact instead.
3. **The discipline.** Nothing clinical is generated. Which drawing and which sentence a diagnosis becomes is decided by a curated library, every skipped line is shown to the clinician, and the whole thing runs with no API keys.

The three things to show, in order

1. The spot searching the body and landing on the jaw.
2. **Not used:** the line it refused to read as a drug.
3. The turn, then the walkthrough read aloud, then Spanish, then print.

If a judge asks one question, it is probably this

"How do you know it will not tell a patient something wrong?"

Because nothing clinical is written by a model. The 205 plain sentences are ours and fixed. What a medicine is for comes from 95 curated class rules keyed on the drug classification, not on the drug's name. Warnings are quoted from the FDA label with a link back to it. A model, when one is configured, only translates or fills a parsing gap, and the demo runs with none. And the clinician reads every sentence before the screen turns, because the screen turning is a physical act they perform.

Numbers we must not get wrong

NUMBER	WHAT IT IS
205	Conditions in the library, each with one plain sentence, the words a real note uses for it, its ICD-10 prefixes, and a drawing with a structure to mark.
21	Categories a patient would look under. Head & brain, Chest & heart, Wrist & hand, Mind & mood, and so on.
18	Hand-drawn anatomical views. Body, head, eye, ear, mouth, neck, shoulder, elbow, wrist and hand, heart, lungs, abdomen, kidneys and bladder, spine, hip, knee, ankle, skin.
9	Step-by-step walkthroughs, each with its own drawing: socket syringe, inhaler with spacer, crutches, eye drops, sling, dressing change, injection pen, ice and elevation, splint care.
95	Curated drug-class rules, keyed on ATC and EPC class identifiers rather than drug names.
~80	Section-header aliases the parser knows, so "Medicine:", "Rx:", "Home medications" and "Discharge Medications" all land in the same place.
8	Note formats in the regression suite, run on every change.
5	Languages: English, Spanish, Vietnamese, Chinese (Simplified), Arabic. Arabic renders right to left.
0	API keys required, including on the published site.
2	Live URLs: aperta.health and aniketh.net/aperta . Both static, both HTTPS.

The stack, in one breath

Next.js 16 with the App Router, TypeScript, Tailwind v4, exported as a fully static site and hosted on GitHub Pages. Drug identity and class from RxNorm and RxClass at the National Library of Medicine. Labels and packaging from openFDA. Translation from MyMemory. Voice from ElevenLabs for the example notes, the browser's own speech for everything else. Drawings made by hand in Procreate.

What each service gives us

SERVICE	WHAT WE ASK IT	KEY
RxNav / RxNorm	What drug is this text, what are its active ingredients, what classes is it in	None
openFDA label	Boxed warning, contraindications, dosing, specific populations, interactions	Optional
openFDA NDC directory	How many marketed generics exist, as a proxy for how hard it is to fill	None
MyMemory	Translation of the patient sentences	None
ElevenLabs	Recorded voice, baked into the build for the example notes	Ours

The retired RxNav interaction endpoint is deliberately not called. It was withdrawn in January 2024, so labelled interactions come from the FDA label text instead.

The demo, click by click

Ninety seconds. Paste the wisdom-tooth note, because it produces a walkthrough, which is the part nobody else has.

0:00	Paste the note.	"This is the note the surgeon already wrote. I am not asking anyone to type anything new."
0:08	Watch the spot search the body and land on the jaw.	"While it reads, the spot is looking for the place."
0:20	Point at Not used .	"It refused to read that line as a drug. Refusing is a feature, and the clinician sees what was refused."
0:32	Scroll to Before you prescribe , pick a medicine.	"Same screen as the list. Duplicates, interactions, the boxed warning, all quoted from the FDA label with a link back."
0:45	Press the button. Turn the laptop.	Say nothing for two seconds. Let them watch the screen turn.
0:52	Body, jaw, back molars. Then the medicine screen.	"One idea per screen, read aloud, and it advances when the reading finishes."
1:05	The socket walkthrough, steps numbered on the drawing.	"This is the phone call three days later that nobody has to make."
1:18	Switch to Spanish .	"Drug names are never translated, because the patient has to match them to the bottle."
1:26	Print this.	"One sheet for the fridge. The drawing goes with it."

Controls worth knowing

- **Space** pauses and resumes. **Left and right arrows** move a screen. The strokes along the top jump to a screen.
- Tap the picture to zoom back out to the whole body.
- **Evergreen** is the dark mode, on only by the button, never by the system. The sun and moon toggle sits in the nav.
- The anatomy library at /diagrams shows all 18 views and every condition, with a one-line explanation of each part on hover.
- Deep links for a clean restart: ?note=wisdom&turn=1, plus &lang=Spanish, &slide=3, &theme=dark.

Second note, if they want another

The broken-wrist discharge summary. It gives two medicines, a splint-care walkthrough, ice and elevation, and a follow-up. Good for showing the medicine sentences and the prescriber check, because ibuprofen and acetaminophen are a real duplicate-risk pair.

How it works, in six steps

1. **Parse.** The note is walked line by line. A header is recognised from about eighty aliases, so "Dx:", "Medicine:", "Discharge Medications" and "Home medications" all land correctly. A line before any header survives only if it unmistakably reads as a drug order. Notes with no headers at all fall through to a prose scan.
2. **Refuse.** Under a medications header, a line becomes a medicine only if it parses as a drug order: a name followed by a number and a unit. Anything else is put in front of the clinician as **Not used** rather than guessed at.
3. **Resolve.** Each medicine goes to RxNorm for its identity, its active ingredients and its classes. A resolution is kept only if the printed name shares a word with the canonical name, or the line carried a dose. This is the guard that stopped "Headache (QOD)" becoming an aspirin and caffeine powder.
4. **Match.** The diagnosis is matched against the 205-condition library. An ICD-10 prefix outranks any phrase, and among phrases the longest match wins, so "ST depression" on an ECG line does not become depression. The match decides the drawing, the structure to mark and the plain sentence.
5. **Build.** What each medicine is for comes from 95 curated class rules keyed on ATC and EPC identifiers, with a visit-context layer where the class alone misleads, and the label's own indication as a last resort. How to take it is rewritten from the sig. Instructions that name a procedure attach one of the nine walkthroughs.
6. **Deliver.** The clinician edits anything, then turns the screen. The story is read aloud, one idea per screen, in the chosen language, and prints on one sheet.

Why the published site needs no server

The parser, the card builder and the prescriber check are ordinary TypeScript with no Node dependencies, so the same functions that run in the API routes of the full app run inside the page on the static site. The build physically moves the API routes aside, and the deployed folder contains none. The page talks to the National Library of Medicine and the FDA directly from the browser.

What the model is allowed to do

Only two things, and only when a key is configured, which the demo does not have. It may fill a parsing gap when the deterministic pass comes back thin, copying text verbatim under a schema that is validated before anything is used. It may translate, under a prompt that forbids adding, softening or reinterpreting a medical statement. It never decides which condition a note is, what a drug is for, or what a warning says.

Product, and the Impiricus track

Why is this an HCP engagement tool and not a patient app?

Because a patient never opens it. The clinician opens it, in the room, from the note they already wrote, and the patient only ever sees a screen that has been turned toward them. Patient education is approved-content territory, adherence is the shared interest between a manufacturer and a prescriber, and this earns the moment by removing work rather than adding a message.

Who pays, and for what?

Not the patient and not the physician. A manufacturer or a platform like Impiricus pays to have approved patient materials delivered at the second a clinician needs them: a spacer walkthrough, an injection-pen walkthrough, a plain-language purpose sentence for their drug. The unit being bought is placement in the explanation, not an impression.

If pushed on ethics: nothing sponsored changes a clinical finding, and the clinician sees every sentence before it is shown to anybody. Content, not recommendations.

Why would a doctor who is already behind use this?

Because it takes the note they have already written and gives back something they would otherwise have to say twice. The interaction is a paste and a button. The printout is the thing that stops the phone call three days later.

How is it measurable if nothing is stored?

Nothing about the patient is stored, and we would keep it that way. What is countable without touching the patient is the material: which walkthrough was turned toward a patient this week, in which language, for which drug class. That is a signal a rep visit cannot produce, and it needs no identifier.

Be honest that this is not built yet. Today there is no analytics of any kind.

The EHR already prints after-visit instructions. Why is this different?

The after-visit summary is the note, reformatted. It is written at a reading level most patients cannot use, it has no picture of where the problem is, it does not read itself aloud, and it does not exist in the patient's language. We start from the same source and produce something a frightened person can repeat back.

Why not just use ChatGPT for this?

Because a general model will always answer, and in this domain a confident wrong sentence is the harm. "Take the headache powder" is a real thing a fuzzy matcher produced for us. Our answer is a curated library, a refusal list the clinician sees, and warnings quoted from the label rather than written. That is what makes it something a health system would let into the room.

What is the wedge? Who is the first customer?

Specialties where the instructions are procedural and the failure is expensive and immediate: oral surgery, orthopaedics, ophthalmology, respiratory. Those are the visits where a walkthrough is the whole value, and they are exactly the visits where a manufacturer has an approved walkthrough already.

How does it get into a clinic? What about the EHR?

Today it is a page with a paste box, which is deliberately the lowest possible bar. The next step is a SMART on FHIR launch inside the chart so the note arrives without a paste. Nothing about the architecture changes, because the note is already just text in and a card out.

What stops this from being an ad?

The clinician is the author of every sentence shown, the screen does not turn until they have seen what was refused, and nothing sponsored can create or alter a finding. Be the answer, not the ad.

Clinical safety and trust

Is this a medical device? Is it regulated?

As built it is patient education, and it is labelled on every screen and every printout as an educational demo and not medical advice. It makes no recommendation and no diagnosis: it restates a diagnosis a clinician already made. A product decision to add anything that recommends would change that answer, which is a reason we have not.

Who is responsible if a sentence is wrong?

The clinician, and the design makes that real rather than nominal. Every sentence is on their screen, editable, before the screen turns. We do not have a path where something reaches a patient that a clinician has not seen.

Can it hallucinate?

Not on the clinical path, because no model writes any of it. The plain sentences are ours and fixed. The purposes are curated and keyed on drug class. The warnings are quoted. With no key configured, which is how the demo runs, no model runs at all.

What if the drug is used off-label?

This is the real weakness of a class-based purpose, and we handle it in two ways. There is a visit-context layer for the cases we know: topiramate for a migraine patient says it makes migraines happen less often, not that it prevents seizures. And every purpose sentence is editable by the clinician before the screen turns. We would want a pharmacist to validate the library before this was used for real.

How do you know the red spot is in the right place?

Every spot is a coordinate on a drawing we made, and every one was checked on a printed sheet that labels each condition beside its drawing. A Colles fracture sits on the lower end of the radius, a torn meniscus on the joint line, wisdom teeth on the back molars. It is our anatomy, so we can put the mark exactly where the injury is, which is something a third-party 3D model cannot do.

What reading level is the patient text?

The plain sentences are written for someone around twelve years old, which is the usual target for patient material, and they describe what happened rather than what to do. Telling the patient what to do is the clinician's line, so instructions are their words, not ours.

Who checks the translation?

Nobody, and we say so. It is free machine translation, which is the right call for a demo and the wrong one for a clinic. A real deployment would use a translator with a data agreement and human-reviewed strings, and the library is short enough that a professional pass over it is affordable.

Does patient information leave the browser?

In English, nothing from the note goes anywhere except drug names, which go to the National Library of Medicine and the FDA to be identified. If a language is chosen, the patient sentences, which include the instructions the clinician wrote, are sent to a public translation service. That is a demo-grade choice and the first thing we would replace. Nothing is stored by us at any point, there are no accounts and no history, and the note is gone when the tab closes.

Never say the word compliant. Say: no data is stored, and here is exactly what is sent where.

Technical

What is deterministic and what is not?

Parsing, condition matching, drug resolution, the purpose sentences, the sig rewriting, the walkthrough detection and the whole prescriber check are deterministic code. The only non-deterministic paths are optional: a model to fill a thin parse or read a photo, and a model as a first choice for translation. The published demo has neither.

How does a diagnosis become a condition?

Each of the 205 conditions carries ICD-10 prefixes and the words a real note uses for it. A code match returns immediately and outranks every phrase. Among phrases the longest match across the whole library wins, matched on word boundaries after punctuation is normalised on both sides. There is also a small guard list for phrases that borrow a condition's name: ST depression on an ECG line is not depression, heartburn is not a burn.

What happens with a note that has no headers?

It falls through to a prose scan that pulls drug orders out of sentences, matches the condition against the whole text, and sorts the remaining sentences into instructions and follow-up by cue words. One of the eight regression cases is exactly this, a free-text history and physical with no structure at all.

How do you identify a drug from free text?

RxNorm's approximate-match endpoint gives ranked candidates, and we walk them until one yields active ingredients, with a second attempt through the concept-history endpoint for drugs that have been retired, which is how a discontinued brand still resolves. Ingredients come back as identifiers, not strings, which is what makes the duplicate check exact.

How does the duplicate-ingredient check actually work?

Set intersection over RxNorm ingredient identifiers, for every pair in the list. No string comparison and no model. Vicodin and Tylenol both contain the same acetaminophen identifier, so the pair is flagged as a duplicate and the same identifier drives a running total against the daily acetaminophen ceiling. Duplicate class is separate, because an ACE inhibitor and an ARB share no code and still should not be taken together, so those come from curated class groups.

A static site with no backend is calling the FDA. How?

Those services allow browser requests and need no key, so the page calls them directly. It means there is no server of ours in the path at all, which is good for privacy and good for a hackathon budget. It also means we inherit the anonymous rate limits, and a throttled request shows up as a section that is not there rather than an error.

How is it tested?

A regression suite of eight note formats runs on every change: a prose history and physical, an EHR export with codes and pipe-delimited sigs, a cardiac discharge with abbreviations, a terse emergency note, a note with junk under Medications, a note using "Medicine:" as its header, and both demo notes. Each case asserts the condition it must resolve to, the medicines that must be found, the strings that must *not* be treated as medicines, and the walkthroughs that must attach.

How would you scale past 205 conditions?

The library is the product, so growth is editorial rather than technical: each entry is one plain sentence, a drawing, a structure, synonyms and codes. The next step we would take is generating candidate entries from a coded terminology and having a clinician approve the sentence, so the human effort is review rather than authorship.

What is cached, and why is a cache committed to the repo?

Every external lookup is cached by its URL, and the recorded audio by a hash of its text. The cache is committed on purpose so a demo on venue wifi does not depend on three public APIs being fast, and so the recorded clips ship with the build instead of being fetched.

What did you build versus what came from a library?

The parser, the condition library, the matcher, the purpose rules, the sig rewriter, the prescriber check, the walkthrough detection, the card builder, the interface and every drawing are ours. We call public data services for drug identity and labels. The drug-identity engine came from an earlier project of ours, PillPile, which found hidden duplicate ingredients by intersecting RxNorm identifiers.

Design, accessibility, and the team

Why hand-drawn anatomy instead of 3D or stock?

Three reasons. A photographic organ reads as gory to someone who has just had bad news. A third-party model cannot be marked with *your* fracture. And a drawing prints, which a 3D viewer does not. We tried 3D first and took it out.

How does one drawing work on both a light and a dark screen?

Each drawing is black ink on transparency, and on the dark board a colour filter turns the same file into beige ink. One asset, two themes, and the portraits on the team page are exempt so faces are never inverted.

What did you do for accessibility?

Text pairings clear the AA contrast standard, the flow works from the keyboard, every drawing carries a description, motion honours the reduced-motion setting, Arabic renders right to left, every screen is read aloud, and the story prints. Type is sized to how much text is on a screen so nothing scrolls and nothing is cut off.

Honest gap: some decorative hairlines and the progress strokes are below the 3:1 mark for non-text contrast, and our custom dropdown does not move focus the way a native one does. Both are on the list.

Why is dark mode off unless you click it?

Because a clinician's laptop is set for the clinician, and the patient is the one who has to read it. A system setting should not decide how a patient's screen looks, so it is a deliberate button. It is called Evergreen, and the light one is Paper.

Who did what?

Two of us. Lauren drew the anatomy, the walkthroughs and the portraits in Procreate, and drove the design decisions about what a frightened person can take in. Aniketh built the parser, the library, the matcher and the prescriber check, and wired the drawings in. The design passes were argued out together, which is why so much got removed.

What was the hardest part?

Making the thing refuse. It is easy to always produce an answer and very hard to decide, in code, that a line is not a drug, and then to show that decision to a clinician in a way that reads as care rather than failure.

What would you do with another week?

Teach-back, where the patient says it back and the tool checks that the three facts that mattered survived. Then form-aware dosing for inhalers, drops and pens, a pharmacist pass over the purpose library, and a launch inside the chart so nothing is pasted.

Where we are weak, and what we say

These are real, verified in our own code. Saying them first is better than being caught by them. The pattern that works: name it, say why it is bounded, say the fix.

1. Dosing sentences assume a pill

The sig rewriter counts puffs, drops and units correctly but prints the word "pill" regardless, so "2 puffs twice a day" reads as "Two pills, twice a day." Patches, sprays and liquid volumes are not handled at all, and durations in weeks are dropped.

Say: "Our dose sentences are pill-shaped today. It counts the dose correctly, but it names the form wrong for inhalers and drops, and that is the next thing we fix, because for an inhaler the form *is* the instruction."

2. Some medicine lines vanish instead of being shown

A drug order with no milligram-style strength, like "Albuterol HFA 2 puffs q4h PRN" or "Latanoprost 1 drop each eye", is not accepted as a medicine, and because it sits under a medications header with no instruction cue it is discarded rather than listed under Not used. Combination strengths written as "Norco 5-325 mg" and percentage strengths like "hydrocortisone 1% cream" also fail to parse.

Say: "The refusal is right, but the reporting is not. Anything we cannot read should appear under Not used, and in that specific shape it does not. It is a small fix and it matters more than it looks, because silence is the one thing this project is not allowed to do."

3. Drug orders written under "Plan" become diagnoses

A note that says "Assessment and Plan: 1. Type 2 diabetes, start metformin 500 mg BID" puts the whole line in the diagnosis list and finds no medicines, because Plan is not treated as a medication section.

Say: "We tuned on discharge summaries and after-visit summaries, where medicines have their own header. A Plan section needs the line split, the diagnosis taken from the first clause and the drug order from the rest. We know the shape of the fix."

4. Negations and rule-outs are not guarded

"No history of stroke" and "rule out appendicitis" are still candidate diagnoses, and only three guard patterns exist for phrases that borrow a condition's name. One of those three is keyed to a condition identifier that does not exist, so it never fires.

Say: "Negation handling is the first thing any clinical parser needs and we have the guard list rather than the guard rule. It is a known gap with a known fix."

5. Measurability and privacy are in tension

The pitch says which walkthroughs were shown is countable. The product stores nothing and has no analytics of any kind.

Say: "Today it is zero storage and zero analytics, which is the honest state. The claim is about what is countable without a patient identifier, and we have not built it. We would rather show you nothing than show you a number we made up."

6. Other things worth conceding quickly

- Translation is free and unreviewed, and the shipped Spanish for the demo notes has at least one bad line. If a judge speaks Spanish, get ahead of it.
- 205 conditions is small. Anything outside it falls back to a whole-body drawing and the clinician's own words.
- Purpose sentences are class-based, so an off-label use can read wrong until the clinician edits it.
- Nothing here has been validated clinically. No patient has used it.
- The condition matcher can use an ICD-10 code, but in the common path the code is stripped from the line before matching, so the synonym usually does the work.

Demo safety

Notes that are certain to work

- The two example notes above the box. Both are prebuilt, including their recorded audio and their Spanish, so they work with no network at all.
- A pasted note in the shape of a discharge summary: a Diagnosis or Dx line, a Medications header with milligram doses, an Instructions list, a Follow-up line.

If a judge wants to paste their own note

Say yes, and set it up in one sentence: "It will work on anything shaped like a discharge summary. If it hands something back as Not used, that is the point of the thing, not a failure." Then watch for an inhaler, eye drops, a patch or a Plan section, because those are the shapes we know are weak, and name the gap before they do.

If something breaks on stage

WHAT YOU SEE	WHAT TO DO
Wifi is gone or slow	Use the two example notes. They are prebuilt and need nothing.
No voice on a pasted note	Expected. Recorded audio ships only for the examples, everything else uses the browser voice. Say that.
A section of the prescriber check is missing	A public API was throttled. Say it reports what it checked, and re-run it on a different medicine.
The card looks thin	Show Not used and explain the refusal. A thin card with a visible refusal is a better demo than a full one.
The page is in a weird state	Reload with a deep link: aperta.health/?note=wisdom&turn=1

Things to never say

NOT THIS	THIS
"It is HIPAA compliant."	"It stores nothing, and here is exactly what is sent where."
"It diagnoses."	"It restates a diagnosis the clinician already made."
"The AI writes the explanation."	"A curated library writes it. A model only translates or fills a parsing gap."
"It is accurate."	"It is deterministic, it is quoted from the label, and every sentence is reviewed by the clinician."
"It is safe."	"It refuses rather than guesses, and it shows you what it refused."

Words that will come up

Sig The directions on a prescription. "PO TID x 7d" is a sig.

PO By mouth.

PRN As needed.

BID, TID, QID Twice, three times, four times a day.

q6h Every six hours. "q4-6h" is a range.

QOD Every other day.

ICD-10 The international code for a diagnosis. S52.502A is a fracture of the lower end of the radius.

RxNorm The National Library of Medicine's naming system for drugs. Gives every drug and ingredient a stable identifier called an RxCUI.

RxClass The companion service that says which classes a drug belongs to.

ATC and EPC Two class systems. ATC is the international therapeutic classification, EPC is the FDA's "established pharmacologic class". Our purpose rules are keyed on these.

NDC The national drug code for a marketed package. Counting them tells you whether a generic exists.

openFDA The FDA's public API for label text and drug listings.

DailyMed Where the full official label lives. Our check links back to it.

Boxed warning The most serious warning on a label, historically called a black box.

Prior authorisation The insurer's permission step before a drug is covered. The full app drafts the letter.

Teach-back Asking a patient to say the plan back in their own words. The evidence-based fix we are automating around.

AVS After-visit summary, the printout a patient leaves with today.

SMART on FHIR The standard way an app launches inside an electronic health record with the chart's context.

Distal radius fracture A broken wrist. A Colles fracture is the common angled version.

Third molar A wisdom tooth.

NSTEMI A type of heart attack.

HNP Herniated disc.

HFA The propellant in a metered-dose inhaler, so "albuterol HFA" is the puffer.

Where things live

THING	FILE
The condition library, 205 entries	<code>lib/anatomy/conditions.ts</code>
The parser	<code>lib/parseNote.ts</code>
Purpose and dose sentences	<code>lib/plainPurpose.ts</code>
The prescriber check	<code>lib/prescribeCheck.ts</code>
Drawings and spot coordinates	<code>lib/anatomy/art.ts</code>
The regression suite	<code>scripts/verify-explain.ts</code>

Run it locally with `npm run dev`. Run the checks with `npm run verify`. Publish with `npm run export:static`.